

Problem 10.12

If the force of interest δ is 0.1, obtain the nominal rate of interest per annum on deposits compounded

- (a) monthly;
- (b) quarterly;
- (c) annually.

Solution

For a constant force of interest,

$$1 + i = e^{\delta}.$$

The equivalent nominal interest rate convertible m times per year satisfies

$$\left(1 + \frac{i^{(m)}}{m}\right)^m = e^{\delta}.$$

Taking the m th root and solving for $i^{(m)}$ gives

$$i^{(m)} = m \left(e^{\delta/m} - 1 \right).$$

Here,

$$\delta = 0.1.$$

Part (a): Monthly compounding.

For $m = 12$,

$$i^{(12)} = 12 \left(e^{0.1/12} - 1 \right).$$

Therefore,

$$\boxed{i^{(12)} = 0.1004178265}.$$

Part (b): Quarterly compounding.

For $m = 4$,

$$i^{(4)} = 4 \left(e^{0.1/4} - 1 \right).$$

Therefore,

$$\boxed{i^{(4)} = 0.1012604821}.$$

Part (c): Annual compounding.

For $m = 1$,

$$i = e^{0.1} - 1.$$

Therefore,

$$\boxed{i = 0.1051709181}.$$